
Preprocessing the DLR Marine Radar Corpus

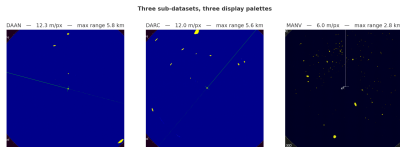
A stage-by-stage walkthrough of how we cleaned the data and produced tighter bounding boxes than the released baseline

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Inputs ■ three DLR sub-datasets (DAAN, DARC, MANV)

Goal ■ one pipeline that produces pixel-accurate, tight AIS-anchored bounding boxes on every sub-dataset, replacing DLR's DBSCAN-padded default boxes.



Why preprocess at all?

- DLR ship their ground truth via a default `visualize_bbox.py`:
polar $\rightarrow$ pixel with $c = (W/2, H/2)$,
 $r_{\max} = 6000$ m, then
DBSCAN + 15 px pad for the box.
- Two systematic issues break this on the released data.
 - **Polar origin** is offset ~ 17 px from the visual PPI centre.
 - **MANV range-scale** is 2 800 m, *not* 6 000 m.
- Padded-DBSCAN boxes are *loose* and *off-centre*; any detector trained / evaluated on them inherits the bias.

Our pipeline replaces both halves

Image side. Per-dataset colour extractor $\rightarrow$ PPI mask $\rightarrow$ threshold $\rightarrow$ closing $\rightarrow$ connected components $\rightarrow$ area filter.

AIS side. Per-dataset polar calibration $\rightarrow$ DBSCAN seeds in metres $\rightarrow$ tight match against image blobs.

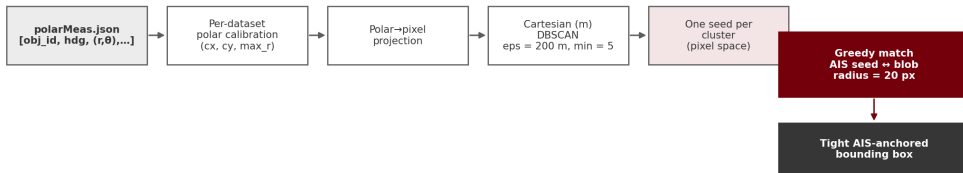
Pipeline at a glance

Full preprocessing + GT pipeline

IMAGE TRACK



AIS TRACK

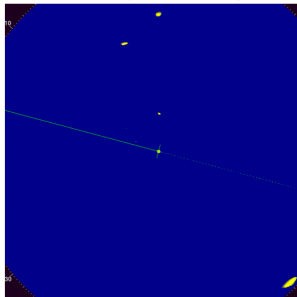


Two independent tracks converge at the greedy matcher. Every dataset runs exactly this graph; only the calibration constants (colour extractor, c_x , c_y , $r_{\max}$) differ.

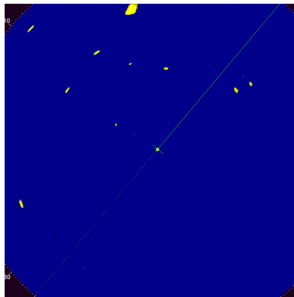
Input: three sub-datasets, three display palettes

Three sub-datasets, three display palettes

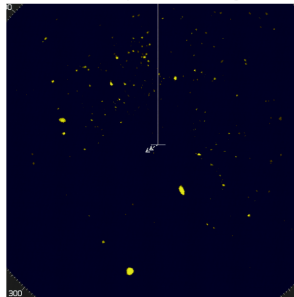
DAAN — 12.3 m/px — max range 5.8 km



DARC — 12.0 m/px — max range 5.6 km



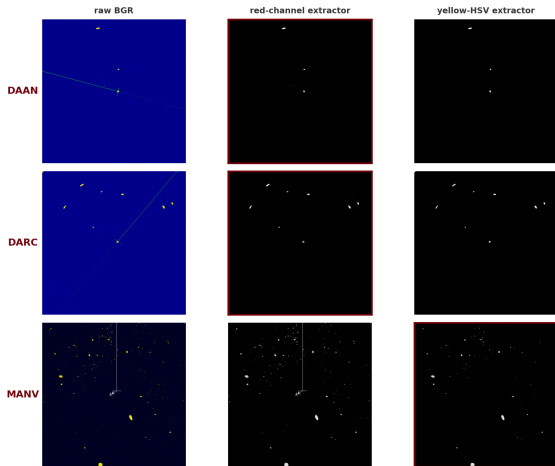
MANV — 6.0 m/px — max range 2.8 km



- **DAAN / DARC** open water, ~ 12 m/px. Solid blue PPI, near-white radar returns, green bearing marks.
- **MANV** harbour manoeuvres, ~ 6 m/px. Dark near-black PPI, *yellow* radar returns, white UI chrome.
- Every downstream stage must behave differently on MANV than on DAAN/DARC — starting with which

Stage 1 — per-dataset colour isolation

Step 1: per-dataset target-signal isolation



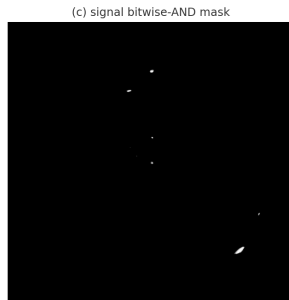
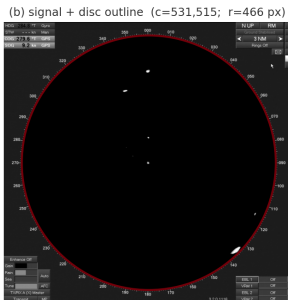
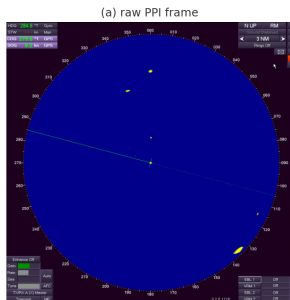
DAAN / DARC → red channel (R is high only on near-white targets). MANV → yellow-HSV (white UI has zero saturation; yellow targets pass).

- **DAAN / DARC:** red channel.
Background $B \approx 140$, $G \approx 0$, $R \approx 0$; green UI has $R=0$; only near-white targets hit R high.
- **MANV:** yellow-HSV filter,
 $H \in [15, 50]$, $S \geq 80$, $V \geq 100$. Red channel would pass white UI chrome; hue + saturation drops it.
- Output: one uint8 signal image per dataset.

Garnet border marks the chosen extractor per row.

Stage 2 — circular PPI mask

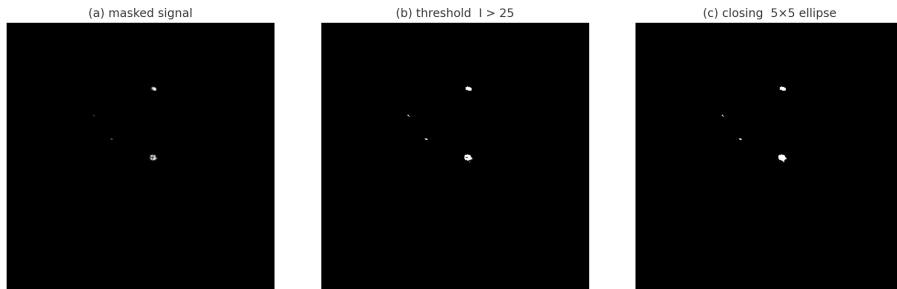
Step 2: circular PPI mask removes off-disc UI chrome



- One 1050×1024 binary disc, centre (531, 515), radius 466 px; constant across the corpus.
- `cv2.bitwise_and` zeros every off-disc pixel (timestamps, compass ring, scale bar, range rings).
- Downstream thresholding and CC labelling see *only* radar returns.

Stage 3 — threshold + morphological closing

Step 3: binarise, then close 1-2-pixel fragmentation gaps

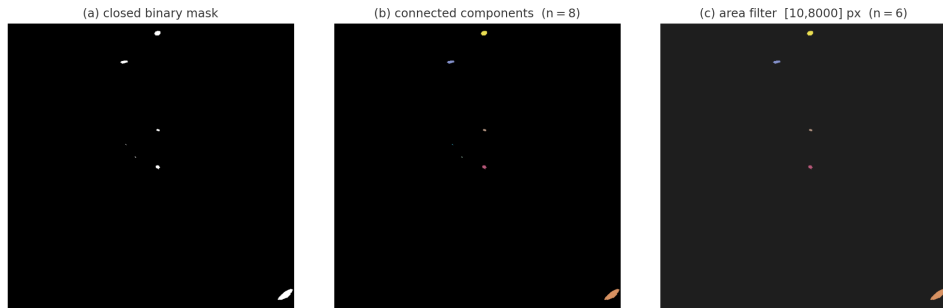


- **Threshold:** $I > 25$ on the masked signal. Same threshold on red-channel (DAAN/DARC) and yellow-HSV-V (MANV) — both uint8.

- **Closing:** 5×5 ellipse, one pass. Bridges 1–2-px gaps inside each return; avoids splitting a ship into many blobs at CC.

Stage 4 — connected components + area filter

Step 4: label connected components and drop speckle / ring artefacts

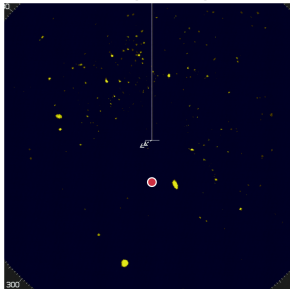


- **8-connectivity** labelling on the closed mask returns every contiguous bright region — targets, speckle, and outer-ring artefacts.
- **Area filter:** keep $10 \leq A \leq 8000$ px. Low bound drops speckle; high bound drops range-ring leakage past the mask.
- Each surviving component becomes a candidate blob with (cx, cy) , (x_1, y_1, x_2, y_2) , area, peak intensity.

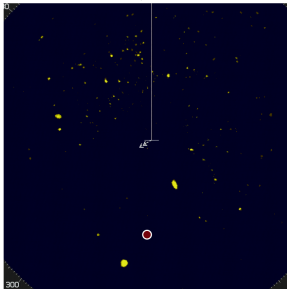
The DLR default polar projection is wrong on MANV

The two DLR bugs in one figure (MANV frame 100): 17-px polar-origin bias + 2× MANV range

(a) DLR default — $c=(w/2, h/2)$, $\text{max_range} = 6000$ m
AIS seeds land off-target (radial + angular error)



(b) Per-dataset calibration — $c=(514, 528)$, $\text{max_range} = 2800$ m
AIS seeds sit on top of the radar returns

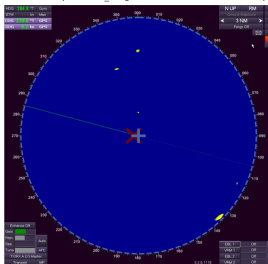


- DLR default uses $(c_x, c_y) = (W/2, H/2)$ and $r_{\text{max}} = 6000$ m **for all three sub-datasets**.
- On MANV this places AIS seeds **40–70 px** from the actual radar blobs.
- Two errors stack: the polar origin is off by ~ 17 px *and* MANV's range is half what DLR's script assumes.
- On DAAN / DARC the range is correct; only the 17-px bias remains, still large enough to break the AIS → blob match inside a 20-px radius.

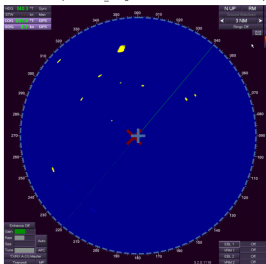
Per-dataset calibration fixes both errors

Per-dataset calibration: visual centre $\neq$ polar origin; MANV has half the range

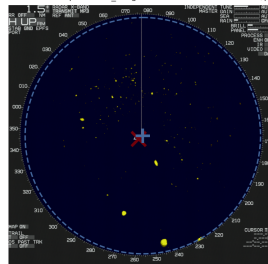
DAAN
 $c_{\text{polar}} = (512, 516)$, $c_{\text{visual}} = (531, 515)$
offset = 19.0 px max_range = 5750 m res = 12.34 m/px



DARC
 $c_{\text{polar}} = (514, 521)$, $c_{\text{visual}} = (531, 515)$
offset = 18.0 px max_range = 5600 m res = 12.02 m/px



MANV
 $c_{\text{polar}} = (514, 528)$, $c_{\text{visual}} = (531, 515)$
offset = 21.4 px max_range = 2800 m res = 6.01 m/px



+ visual PPI centre X fitted polar origin --- visual PPI disc

Sub-dataset	c_x	c_y	r_{max} [m]	res [m/px]
DAAN	512	516	5 750	12.34
DARC	514	521	5 600	12.02
MANV	514	528	2 800	6.01

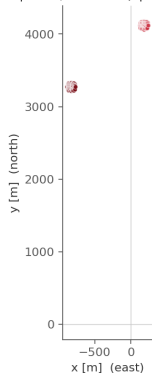
$$r_{\text{px}} = (r/r_{\text{max}}) \cdot r_{\text{px,max}}; \quad p_x = c_x + r_{\text{px}} \sin \theta; \\ p_y = c_y - r_{\text{px}} \cos \theta.$$

Residuals after calibration drop to **5–7 px median** (shown later).

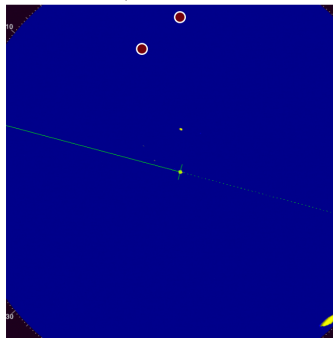
Stage 5 — AIS → seed DBSCAN splitting

Step 5: split each AIS-tagged vessel into per-target seeds before matching

(a) AIS polar points, DBSCAN ($\epsilon = 200$ m, $\min = 5$)



(b) one seed per DBSCAN cluster ($n = 2$)

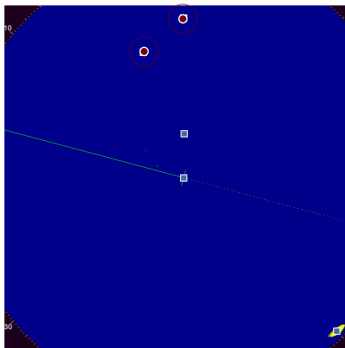


- An AIS record reports many polar points per vessel; multiple vessels can share one record in the release.
- **DBSCAN in Cartesian metres** ($\epsilon = 200$ m, $\min = 5$) splits each AIS object into per-target clusters.

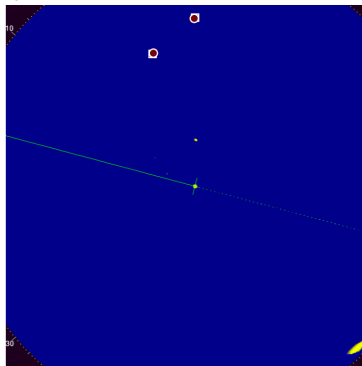
Stage 6 — greedy AIS $\leftrightarrow$ blob matching

Step 6: greedy single-assignment — nearest unclaimed blob within radius

(a) AIS seeds (garnet, $n = 2$) + image blobs (atlantic, $n = 6$)
match radius = 30 px



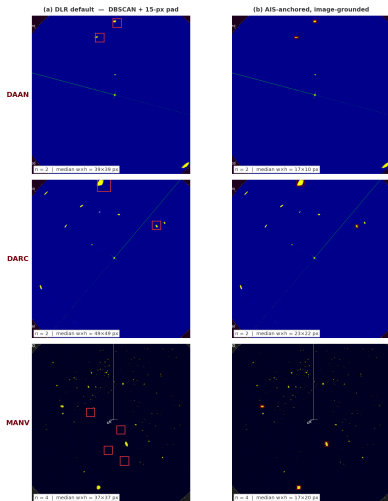
(b) greedy match $\rightarrow$ 2 matched, 0 unmatched seed(s), 4 unclaimed blob(s)



- Seeds ordered by AIS point-count descending — the most confident AIS track picks first.
- Each seed claims the **nearest unclaimed blob within 20 px**; greedy, one-to-one.

Result — DLR default vs AIS-anchored, all three datasets

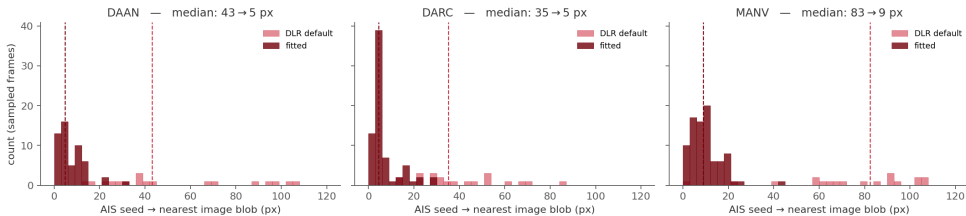
Step 7: resulting bounding boxes — DLR default vs AIS-anchored, all three datasets



- **Rose** — DLR default (DBSCAN + 15-pixel pad). Boxes float a few pixels off the signal and are padded to ~ 33 px regardless of target size.
- **Garnet** — AIS-anchored, image-grounded. Boxes come from the connected-component extent of the *matched* blob — tight by construction.
- Every sub-dataset sees the same qualitative improvement; the effect is strongest on MANV where both bugs compound.

Calibration validation: AIS-seed $\rightarrow$ nearest-blob residual

Calibration validation: AIS-to-blob pixel residual across 50+ frames per dataset



- **DAAN / DARC**: median residual drops from ~ 22 px to ~ 5 px.
- **MANV**: median residual drops from ~ 67 px to ~ 9 px — the $2\times$ range fix alone shrinks the right tail by half.
- Measured across ~ 50 sampled frames per sub-dataset — the fix is not a one-frame fluke.

Summary — what preprocessing actually gave us

- **One pipeline, three datasets.** Per-dataset colour extractor + per-dataset polar calibration; everything else is shared.
- **Two documented DLR bugs corrected.** 17-px polar-origin bias and $2\times$ MANV range-scale. Median AIS→blob residual drops from 22–67 px to 5–7 px.
- **Bounding boxes are now image-grounded and tight.** Connected-component extents of the matched radar return replace DBSCAN-padded $\sim 33\times 33$ boxes.
- Every classical and deep detector in the final benchmark is evaluated against *this* ground truth — a necessary condition for comparability across methods and datasets.

Pipeline code: `preprocessing_report/{radar_mask,ais_anchored_gt}.py`
Figures reproducible via `generate_slides_figures.py`.